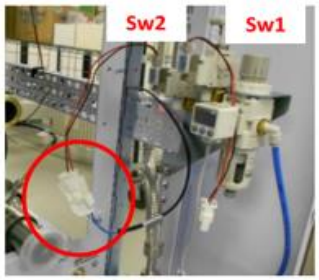
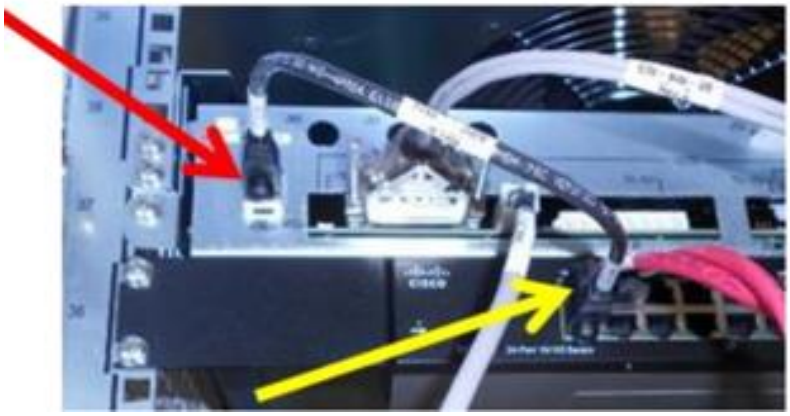
**Build Overview:**

1 **CDU T-tubes Install**
2 **Air Regulator Install**
3 **Label Install**



**CHANGE HISTORY**

| Date | BPM Rev | Change Details | Updated by |
|---|---|---|---|
| | 1 | New Issue | JOSEPH AMAD |

**Prepared By:**
JOSEPH R. AMAD
Manufacturing Engineer

**Checked By:**
IVAN NOWELL PANIQUE
Team Lead

**Approved By:**
SHERWIN ARIOSTE
Operations Manager

**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:**

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 633-799-00 | ELEC-MECH, UF2 24SLOT SUPPORT CABINET INTEGRATION | 1 |

**MACHINE:** N/A

METHOD: 1. Scope of application

This BPM is suitable for UF Plus Support Cabinet Assembly
Applicable Support Cabinet Model 633-799-00
The sub-part model in Support Cabinet are 629-502-01 ADB CHASIS SUB ASSY, 653-315-00 CDBU SUB ASSY and 630-945-10 GEMINI SMC TRAY

MATERIAL:

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 472-426-20 | PAN HEAD CROSS RECESSED SCREW M5X0.8 x 20MM L | 4 |
| 2 | 472-426-12 | PAN HEAD CROSS RECESSED SCREW M5X0.8 x 12MM L | 2 |
| 1 | 631-560-00 | CABLE UF2 ENET SW TO CDU TELECOM WITH CONN | 1 |
| 1 | 634-442-00 | CABLE CDU EMO W/ CONN | 1 |
| 2 | 634-437-00 | CABLE GROUND CABLE CDU W/ CONN | 1 |
| 3 | 603-096-03 | NUT HEX M6x10Wx5MMH W/TOOTHWSHR SS | 1 |
| 1 | 813-929-01 | HDW VELCRO STRP 5/8 | 3 |
| 1 | 361-882-00 | HDW 1.5" SANITRY CLAMP 2PIN SS | 4 |
| 2 | 640-247-00 | CAST STEEL OR IRON, BALL END SPACER, MISC. | 4 |
| 3 | 470-778-04 | WASHER SPLT LOCK 5/16, GR.8, .322IDx.583ODx.078"THK STL.ZN. | 4 |
| 4 | 470-022-85 | NUT, HEX 5/16-18 THD. 1/2"WX17/64"H GR.5 STEEL ZN. PLTD. | 4 |
| 1 | 633-220-00 | STEEL FLEXIBLE TUBING CDU PSU PORT ADAPTER WITH FITTINGS | 2 |
| 2 | 360-557-08 | PRESS FIT STRAIGHT FITTING 5/8" TUBE ID x SAE 3/4-16 W/EPDM | 2 |
| 3 | 638-030-00 | Molykote Grease | 1 |
| 1 | 361-883-00 | SANITRY LBOW 1.5 OD 90DG 304SS | 2 |
| 2 | 361-882-00 | HDW 1.5" SANITRY CLAMP 2PIN SS | 4 |
| 3 | 361-886-00 | GASKT SANITRY CLAM 1.5"F EPDM | 4 |
| 1 | 634-569-00 | FORGED OR STAMPED STEEL AIR FLOW REG MNT BKT MISC | 1 |
| 2 | 614-766-90 | FASTNR RP-01701 12X30X10MM HT. M8 THRD. HOLE ZN.ALLOY | 2 |
| 3 | 470-933-01 | SCREW CAP SKT BUTHD M8x1.25 20MML SS | 2 |
| 4 | 470-502-07 | WASHER LK SPLT M8 14.8MM OD SS | 2 |
| 5 | 470-501-06 | WASHER FLAT 8.4MM ID 16MM OD SS | 2 |
| 1 | 637-027-00 | ELECTRO MECHANICAL AIR REGULATOR FOR SEMI TEST | 1 |
| 1 | 637-717-00 | ELECTRO MECHANICAL DIXON FITTING TO 6MM TUBE FOR SEMI | 1 |
| 2 | 637-746-01 | PLASTIC, TUBE 6MM ODX1MMX560MM, BLUE, HOSE , PIPE | 1 |
| 1 | 634-441-00 | CABLE CDU INTERLOCK SOLENOID W/CONN | 1 |
| 1 | 637-746-00 | PLASTIC, TUBE 6MM ODX1MMX500MM, BLUE, HOSE , PIPE | 1 |
| 1 | 479-133-22 | LABEL, LINE VLT PRSN | 1 |
| 2 | 638-197-01 | LABEL WARNING T-6.00X2.00 0.063 CORNER RADIUS NO HOLES | 1 |
| 1 | 479-133-18 | LBL DANGER HAZ VLT 2.75X1.35" | 1 |
| 1 | 479-133-15 | LABEL, CHASSIS GROUN | 1 |

**MACHINE:**

| NO. | DESCRIPTION | PARAMETER |
|---|---|---|
| 1 | PH2 BIT | 3.3 N.m |
| 1 | 10MM SOCKET HEAD | 5.6 N.m |
| 1 | 13MM SOCKET HEAD | 3.3 N.m |
| 2 | 1INCH OPEN END WRENCH | 1 1/4 TURN |
| 1 | 17MM SOCKET HEAD | 5.6 N.m |
| 1 | CABINET LEG TOOL | N/A |
| 2 | IRON FLAT PLATE | N/A |

**METHOD:**

Flow chart of operation steps

| | |
|---|---|
| BPM preparation | See (1. Scope of application) |
| ↓ | |
| Preparation Tools | See (2. Tools/Materials/Parameters---1. Tools) |
| ↓ | |
| Prepare Materials | See (2. Tools/Materials/Parameters---2. Materials) |
| ↓ | |
| Complete Operation | See (3. Operation Steps) |
| ↓ | |
| Inventory | See (2. Tools/Materials/Parameters---2. Materials) |
| ↓ | |
| Operation Check | See (4. Matters needing attention) |
| ↓ | |
| Inventory collection tools | See (2. Tools/Materials/Parameters---1. Tools) |
| ↓ | |
| Inventory collections tools | See (2. Tools/Materials/Parameters---1. Tools) |

**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:**

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 634-442-00 | CABLE CDU EMO W/CONN | 1 |
| 2 | 634-437-00 | CABLE GROUND CABLE CDU W/CONN | 1 |
| 3 | 603-096-03 | NUT HEX M6x10Wx5MM W/TOOTHWSHR SS | 1 |

**MACHINE:**

| NO. | DESCRIPTION | PARAMETER |
|---|---|---|
| 1 | 1OMM SOCKET HEAD | 5.6 N.m |

**METHOD:**

2.1 Take a wire (634-442-00) as shown in the figure, connect one end to the ADB-P69 interface, and the other end to the CDU-J2 interface.

2.2 Take a wire (634-437-00) as shown in the figure, connect one end to the ground panel Mecca (installed in the previous step), and connect the other end to the CDU ground port.

Use one M6 nut (603-096-03), the tool is a 10mm socket wrench, and the torque value is 5.6 N·m.

**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:**

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 813-929-01 | HDW VELCRO STRP 5/8 | 2 |

**MACHINE:**

**METHOD:**

2.3 Insert the CDU power cord into the J10 interface as shown in the figure. Press the power cord in and turn it clockwise to screw in

2.4 Tie the wire to the guide rail with 2 cable ties (813-929-01) as shown in the figure.

**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:**

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 361-882-00 | HDW 1.5" SANITRY CLAM 2PIN SS | 4 |
| 2 | 640-247-00 | CAST STEEL OR IRON, BALL END SPACER, MISC | 4 |
| 3 | 470-778-04 | WASHER SPLT LOCK 5/16, GR.8, .322IDx.583x.078"THK STL.ZN. | 4 |
| 4 | 470-022-85 | NUT, HEX 5/16-18 THD. 1/2"Wx17/64"H GR.5 STEEL ZN. PLTD. | 4 |

**MACHINE:** N/A

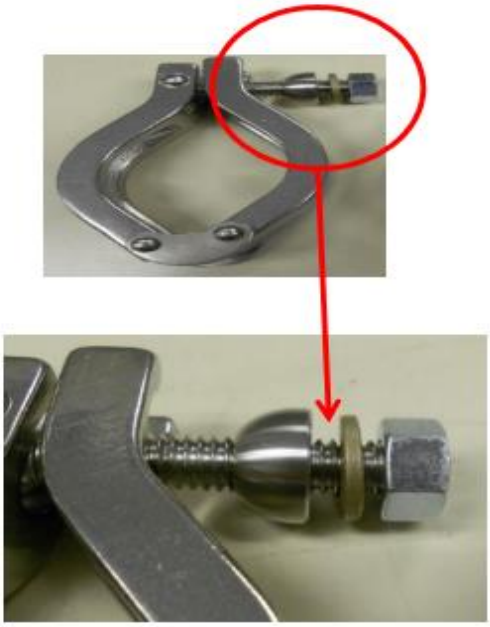
**METHOD:**

3.1 Reconfigure 4 clips (361-882-00) as shown in the figure:

a. Remove and discard the nut.

b. Load the following materials into the screw:

1. 640-247-00 ball head gasket.
2. 470-778-04 spring cushion.
3. 470-022-85 hex nut.



**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:**

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 633-220-00 | STEEL FLEXIBLE TUBING CDU PSU PORT ADAPTER WITH FITTINGS | 2 |
| 2 | 360-557-08 | PRESS FIT STRAIGHT FITTING 5/8" TUBE ID x SAE 3/4-16 W/EPDM | 2 |

**MACHINE:**

| NO. | DESCRIPTION | PARAMETER |
|---|---|---|
| 1 | 15/16 OPEN END TORQUE WRENCH | 6.8 N.m |

**METHOD:**

3.2 Take 2 T-tubes "633-220-00", 2 pipe fittings (360-557-08) and Molykote grease (638-030-00) as shown in the picture.

a. Apply Molykote grease to the O-ring of the pipe fitting.

b. Put the pipe into the T-shaped pipe and fix it with a 15/16 open torque wrench. The torque value is 6.8 N·m.

c. Repeat the above operation on the second T-tube.

**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:**

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 361-883-00 | SANITRY LBOW 1.5 OD 90DG 304SS | 1 |
| 2 | 361-882-00 | HDW 1.5" SANITRY CLAMP 2PIN SS | 2 |
| 3 | 361-886-00 | GASKT SANITRY CLAMP 1.5"F EPDM | 2 |

**MACHINE:** N/A

**METHOD:**

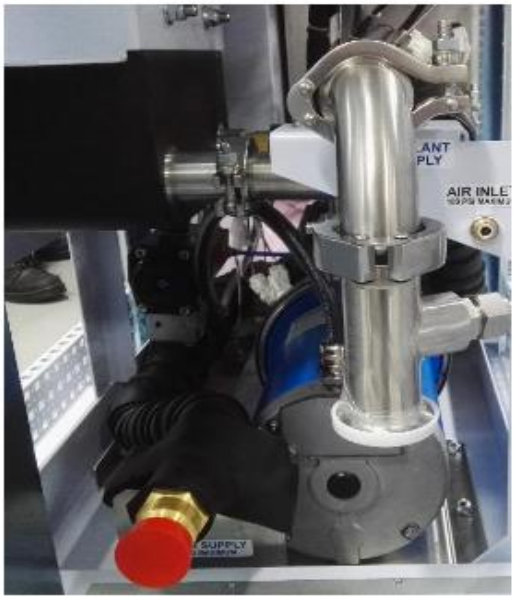
3.3 Install the 90° elbow (361-883-00) on the CDU cooling supply pipe as shown in the figure. Use 1 clip (361-882-00) and washer (361-886-00) fixed.

Note: The outlet direction of the outlet 90° elbow is downward as shown in the figure. The direction of the nut on the clamp is toward the upper right position.

3.4 Install the T-shaped pipe (633-220-00) on the previous 90° elbow as shown in the figure. Use 1 clip (361-882-00) and washer (361-886-00) fixed.

Note: The exit direction of the T-tube is to the right as shown in the figure. The direction of the clamp nut is toward the left position on the back.

Note: Do not use torque to fix the clip. The T-shaped tube needs to be loosened when the tube is installed in the T-shaped fitting.



Example orientation of clamp (Coolant Supply Side)

**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:** N/A

**MACHINE:**

| NO. | DESCRIPTION | PARAMETER |
|---|---|---|
| 1 | 13MM SOCKET HEAD | 3.3 N.m |
| 2 | 1INCH OPEN END WRENCH | 1 1/4 turn |

**METHOD:**

3.5 Insert the hose into the T-tube as shown in the figure.

a. Rotate the 90° elbow and T-tube direction as required so that the hose can be inserted into the pipe fitting.

b. Confirm that the hose is fully inserted into the fitting.3.6 Manually tighten the nuts on the pipe fittings.

3.6 Manually tighten the nuts on the pipe fittings.

3.7 Use a 13mm socket wrench to fix the nut on the clamp with a torque value of 3.3 N·m.

3.8 Mark the nuts on T-tubes and fittings.

3.9 Use a 1-inch open-end wrench to fix the nut on the pipe fitting, and twist the nut to the 1-1/4 position.

-Swagelok fastening process —

1) Insert the hose fully into the T-tube and tighten the nut by hand.

2) Mark the nuts on T-tubes and fittings

3) Use a 1-inch open-end wrench to fix the nut on the pipe fitting, and twist the nut to the 1-1/4 position.

**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:**

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 361-883-00 | SANITRY LBOW 1.5 OD 90DG 304SS | 1 |
| 2 | 361-882-00 | HDW 1.5" SANITRY CLAMP 2PIN SS | 2 |
| 3 | 361-886-00 | GASKT SANITRY CLAMP 1.5"F EPDM | 2 |

**MACHINE:** N/A

**METHOD:**

3.10 Install the 90° elbow (361-883-00) on the CDU cooling return pipe as shown in the figure. Use 1 clip (361-882-00) and washer (361-886-00) fixed.

Note: The outlet direction of the outlet 90° elbow is downward as shown in the figure. The direction of the nut on the clamp is toward the lower right position.

3.11 Install the T-tube (633-220-00) on the previous 90° elbow as shown in the figure. Use 1 clip (361-882-00) and washer (361-886-00) fixed.

Note: The exit direction of the T-tube is to the right as shown in the figure. The direction of the clamp nut is toward the left position on the back.

Note: Do not use torque to fix the clip. The T-shaped tube needs to be loosened when the tube is installed in the T-shaped fitting.

**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:** N/A

**MACHINE:**

| NO. | DESCRIPTION | PARAMETER |
|---|---|---|
| 1 | 13MM SOCKET HEAD | 3.3 N.m |
| 2 | 1INCH OPEN END WRENCH | 1 1/4 turn |

**METHOD:**

3.12 Insert the hose into the T-tube as shown in the figure.

a. Rotate the 90° elbow and T-tube direction as required so that the hose can be inserted into the pipe fitting.

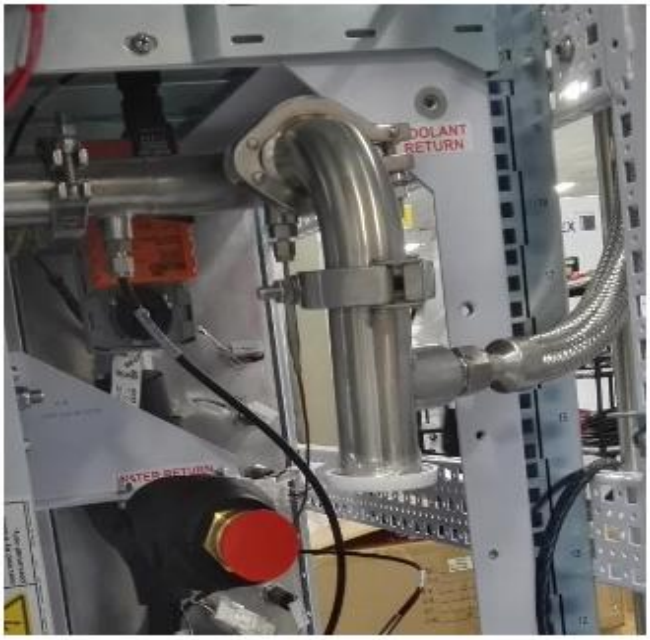
b. Confirm that the hose is fully inserted into the fitting.3.6 Manually tighten the nuts on the pipe fittings.

3.13 Manually tighten the nuts on the pipe fittings.

3.14 Use a 13mm socket wrench to fix the nut on the clamp with a torque value of 3.3 N·m.

3.15 Mark the nuts on T-tubes and fittings.

3.16 Use a 1-inch open-end wrench to fix the nut on the pipe fitting, and twist the nut to the 1-1/4 position.




1) Insert the hose fully into the T-tube and tighten the nut by hand.
2) Mark the nuts on T-tubes and fittings
3) Use a 1-inch open-end wrench to fix the nut on the pipe fitting, and twist the nut to the 1-1/4 position.

**MAN:** :CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL: N/A**

**MACHINE: N/A**

**METHODS:**

Reassemble the joint sleeve. (Use this page for rework or secondary assembly of joint casing)

You may disassemble or reassemble the joint casing several times. Please use this page to install.

Note: Before removing the joint casing part, it must be ensured that the system has been decompressed.

Before disassembly, make a mark and draw a line on the nut and sleeve joint at the back of the pipe.

The purpose of making these marks is to ensure that the nuts can be returned to their original positions when they are reassembled and screwed back.

Insert the pipe with the pre-installed metal ferrule into the sleeve joint until the front metal ferrule is against the joint.

While using a wrench to clamp the casing joint, use another wrench to rotate the nut to the previously tightened position, keeping it flush with the mark made before. At the same time, you will feel the resistance increase significantly, tighten the nut gently.

**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:**

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 690-771-00 | PLASTIC, INTRUCTION, REGULATOR, PRESSURE, LABEL | 1 |

**MACHINE:** N/A

**METHOD:** Installation Air Regulator Label

1. From **Maintanance Side** of Cabinet, at the circled area paste 1pcs Label (690-771-00)

2. Before paste the label, make sure the label align as shown in diagram.

**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:**

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 634-569-00 | FORGED OR STAMPED STEEL AIR FLOW REG MNT BKT MISC | 1 |
| 2 | 614-766-90 | FASTNR RP-017-1 12X30X10MM HT. M8 THRD. HOLE ZN. ALLOY | 2 |
| 3 | 470-933-01 | SCREW CAP SKT BUTHD M8.x1.25 20MML SS | 2 |
| 4 | 470-502-07 | WASHER LK SPLT M8 14.8MM OD SS | 2 |
| 5 | 470-501-06 | WASHER FLAT 8.4MM ID 16MM OD SS | 2 |

**MACHINE:**

| NO. | DESCRIPTION | PARAMETER |
|---|---|---|
| 1 | 5MM HEX BIT | 13.5 N.m |

**METHOD:**

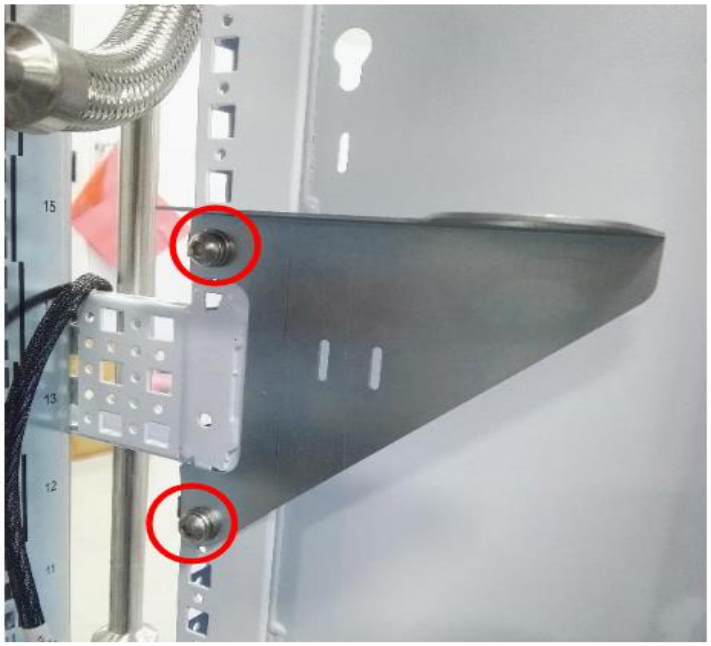
4.1 As shown in the figure, install a bracket (634-569-00) to the frame at the lower right corner of the cabinet maintenance surface. Use 2 M8 frame locking blocks (614-766-90)

And the following materials, the tool is a 5mm hexagon socket head, the torque value is 13.5 N·m.

a. 2 M8 screws (470-933-01).

b. 2 M8 spring pads (470-502-07).

c. 2 M8 flat washers (470-501-06).



**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:**

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 637-027-00 | ELECTRO MECHANICAL AIR REGULATOR FOR SEMI TEST | 1 |

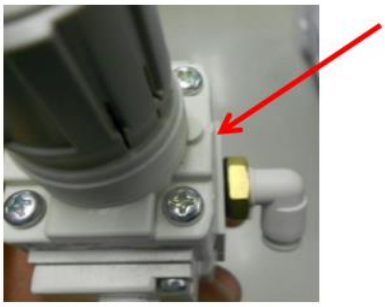
**MACHINE:**

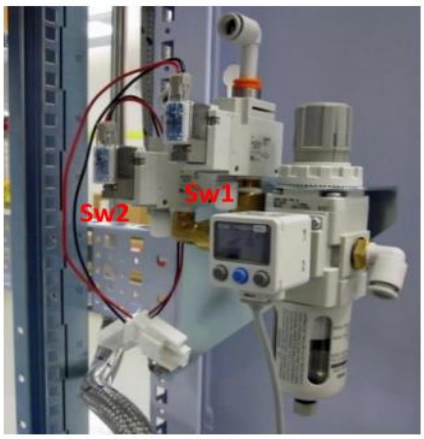
| NO. | DESCRIPTION | PARAMETER |
|---|---|---|
| 1 | AIR REGULATOR SOCKET | 2.0 N.m |

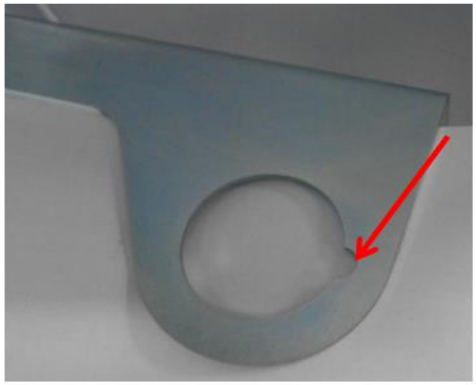
**METHOD:**

4.2 Remove the plastic nut on the regulator (637-027-00) as shown in the figure.

4.3 Install the regulator on the bracket as shown in the figure-when installing the locating PIN on the regulator into the notch of the bracket-install from below and fix it with a plastic nut.





**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:**

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 637-717-00 | ELECTRO MECHANICAL DIXON FITTING TO 6MM TUBE FOR SEMI | 1 |
| 2 | 637-746-01 | PLASTIC, TUBE 6MM ODX1MMX560MM, BLUE, HOSE, PIPE | 1 |

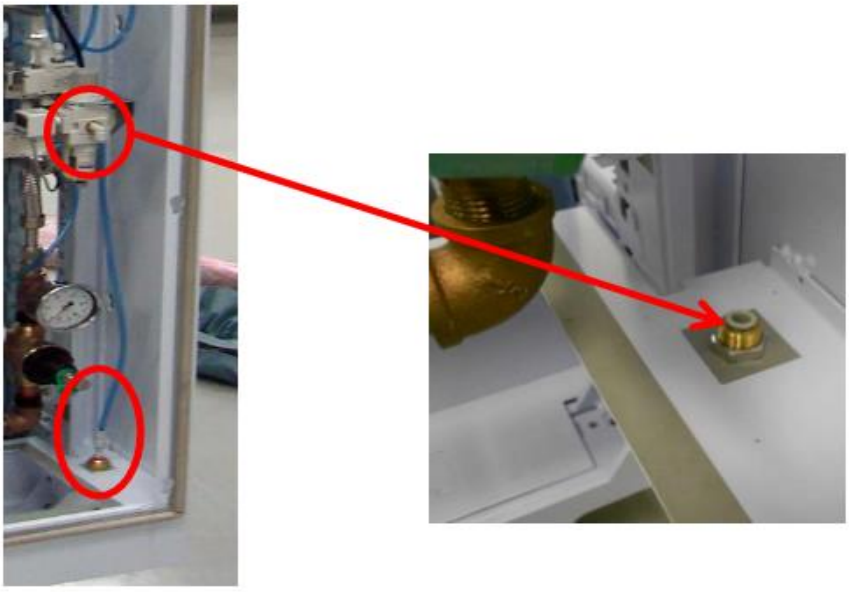
**MACHINE:**

| NO. | DESCRIPTION | PARAMETER |
|---|---|---|
| 1 | 17MM SOCKET HEAD | 5.6 N.m |

**METHOD:**

4.4 Install 1 Dixon connector (637-717-00) under the air conditioner at the bottom of the cabinet as shown in the figure. The tool used is a 17mm socket wrench, The torque value is 5.6 N·m.

4.5 As shown in the figure, install a 560mm air pipe (637-746-01) to the input end of the regulator, and connect the other end to the Dixon connector.



**MAN:** CERTIFIED OPERATOR, TECHNICIAN

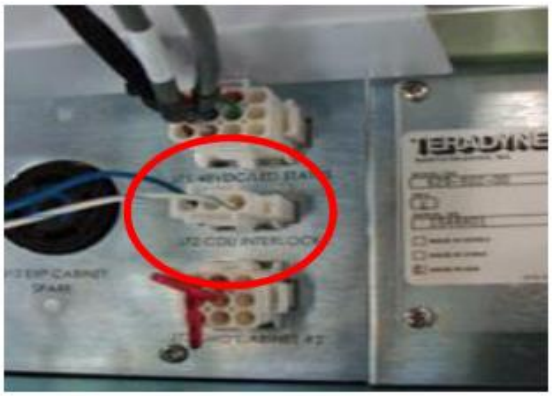
**MATERIAL:**

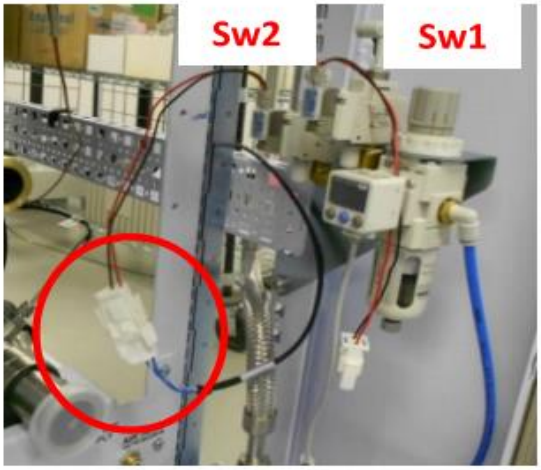
| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 634-441-00 | CABLE CDU INTERLOCK SOLENOID W/CONN | 1 |
| 2 | 813-929-01 | HDW VELCRO STRP 5/8 | 1 |

**MACHINE:** N/A

**METHOD:**

2.5 Take a CDU interlocking cable (634-441-00) as shown in the figure, connect one end to the ADB-P72 interface, and the other end to the air conditioner SW2 interface. Place the extra wire on the frame rail and fix it with a cable tie (813-929-01).





**MAN:** CERTIFIED OPERATOR, TECHNICIAN

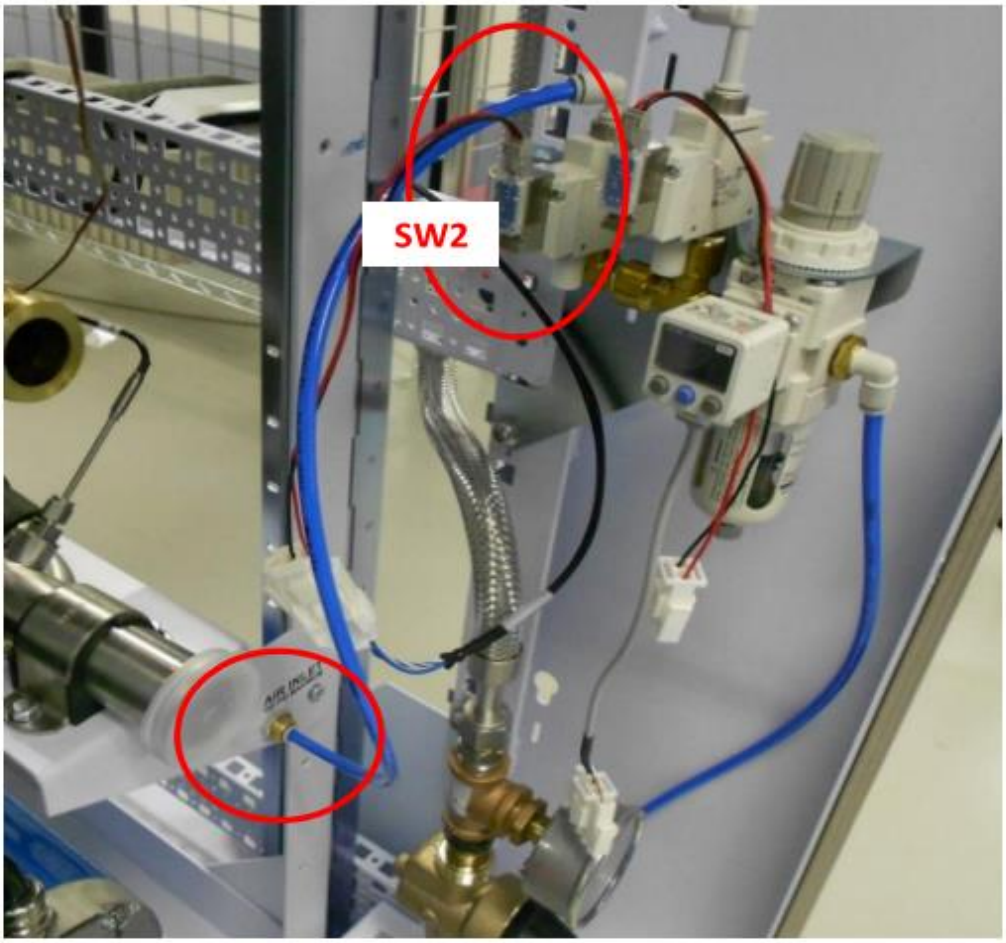
**MATERIAL:**

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 637-746-00 | PLASTIC, TUBE 6MM ODX1MMX500MM, BLUE, HOSE, PIPE | 1 |

**MACHINE:** N/A

**METHOD:**

2.6 Insert a 500mm air pipe (637-746-00) into the outlet of the SW2 air regulator as shown in the picture, and insert the other end into the CDU port.



**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:**

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 479-133-22 | LABEL, LINE VLT PRSN | 1 |
| 2 | 638-197-01 | LABEL WARNING T-6.00X2.00 0.063 CORNER RADIUS NO HOLES | 1 |

**MACHINE:** N/A

**METHOD:**

3.1 Install a "Line Voltage Present" label (479-133-22) on the main circuit breaker panel as shown in the figure. Put the top of the label Keep flush with the top of the circuit breaker and stick to the center of the panel

3.2 Install a warning label (638-197-01) on the front panel of the power rack as shown in the figure. Using Sticker Template

**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:**

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 704-921-00 | CABLE UF2 ENET SW TO CDU TELECOM WITH CONN | 1 |

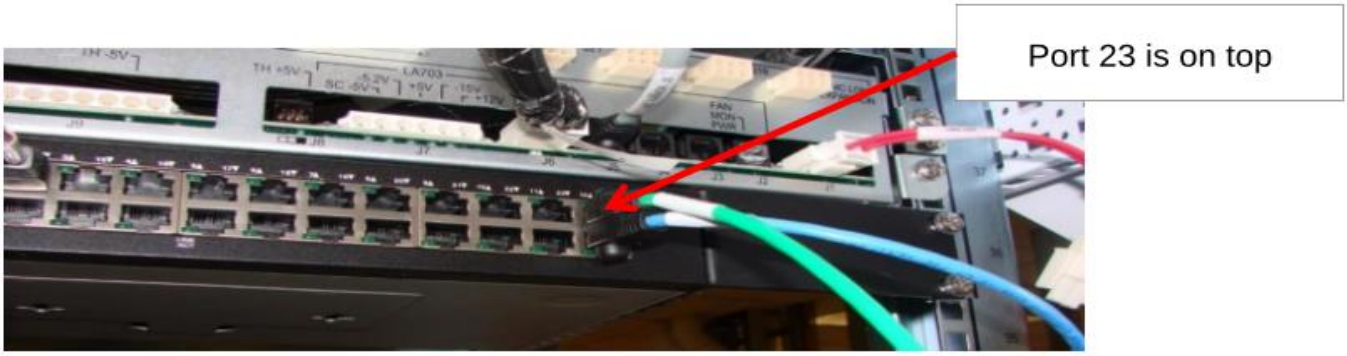
**MACHINE:** N/A

**METHOD:**

1.3 Take a wire (704-921-00), one end is connected to Ethernet switch 23 port, and the other end is connected to CDU-J5 port.

a. Insert the line end label "SC 10/100 ENET SWITCH" into port 23 of the Ethernet switch.

b. Hang the cable through the opening slot of the power supply rack and connect it to the J5 interface of the CDU.



**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:**

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 704-922-01 | CABLE UF2 SMC ENET PNL TO SW SMC TELECOM WITH CONN | 1 |

**MACHINE:** N/A

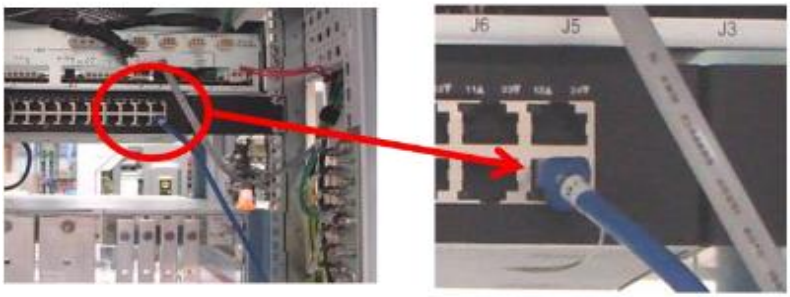
**METHOD:**

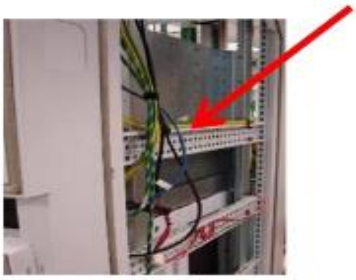
7.1 Take a wire (704-922-01), one end is connected to Ethernet switch port 22, and the other end is connected to the back of theinterface panel of the operation surface "Enet from Computer" port.

a. Insert the connector labeled "SC 10/100 ENET SWITCH" into the port 22 of the Ethernet switch as shown in the figure.

b. As shown in the figure, place the wire outside the cabinet and vertically to the guide rail.

c. As shown in the figure, insert the connector labeled "SC COMP INTFPNL P3" into the back of the interface panel of the operation surface "Enet from Computer" port.



**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:**

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 704-920-00 | CABLE UF2 ENET SW TO PSU TELECOM WITH CONN | 2 |
| 2 | 704-919-00 | CABLE UF2 ENET SW TO SMC TELECOM WITH CONN | 1 |

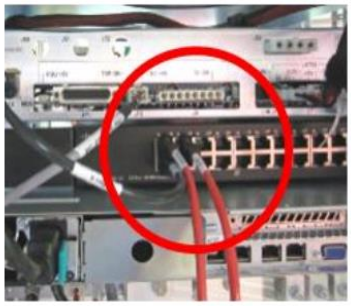
**MACHINE:** N/A

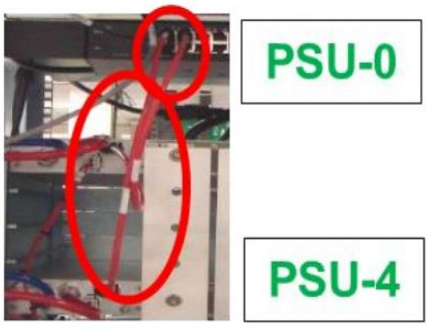
**METHOD:**

7.2 Take 2 wires (704-920-00), connect one end to the Ethernet switch, and the other end to the PSU.

a. Install the cable, one end is connected to Ethernet switch port 1, and the other end is connected to PSU#0.

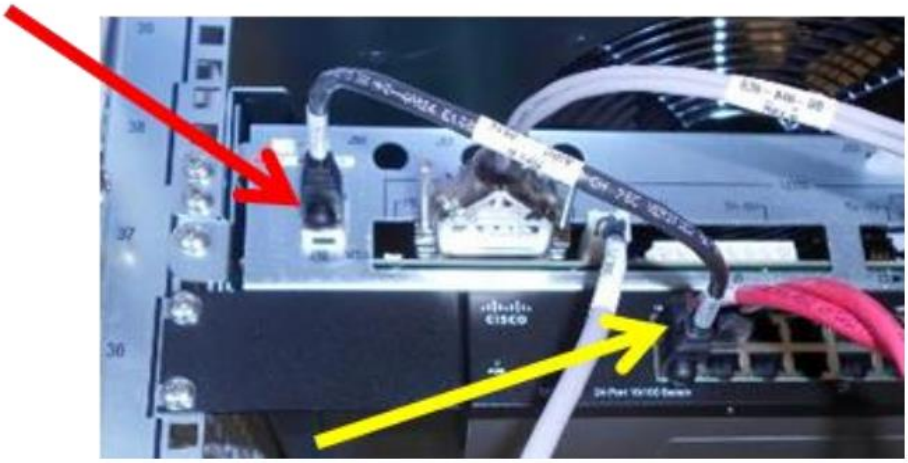
b. Install the cable, one end is connected to Ethernet switch port 8,and the other end is connected to PSU#4.






7.3 Take a wire (704-919-00), connect one end to the Ethernet switch, and the other end to the SMC.

a. Install the cable, one end is connected to Ethernet switch port 5, and the other end is connected to SMC-J12.



**MAN:** CERTIFIED OPERATOR, TECHNICIAN

**MATERIAL:**

| ITEM | P/N | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | 479-133-15 | LABEL, CHASSIS GROUN | 4 |

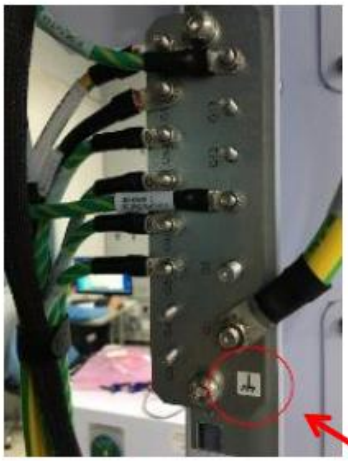
**MACHINE:** N/A

**METHOD:**

3.5 Put a grounding label (479-133-15) on the 4 positions as shown in the figure.

i. One piece on the left side of the grounding snap ring in the upper right corner of the machine maintenance surface, one piece on the left

ii. One piece on the left side of the grounding snap ring in the upper right corner of the machine operation surface



Maintenance Surface

Grounding Snap ring